

Task 1:

RJ-45 Pin Layout & T568A vs T568B Wiring Standards

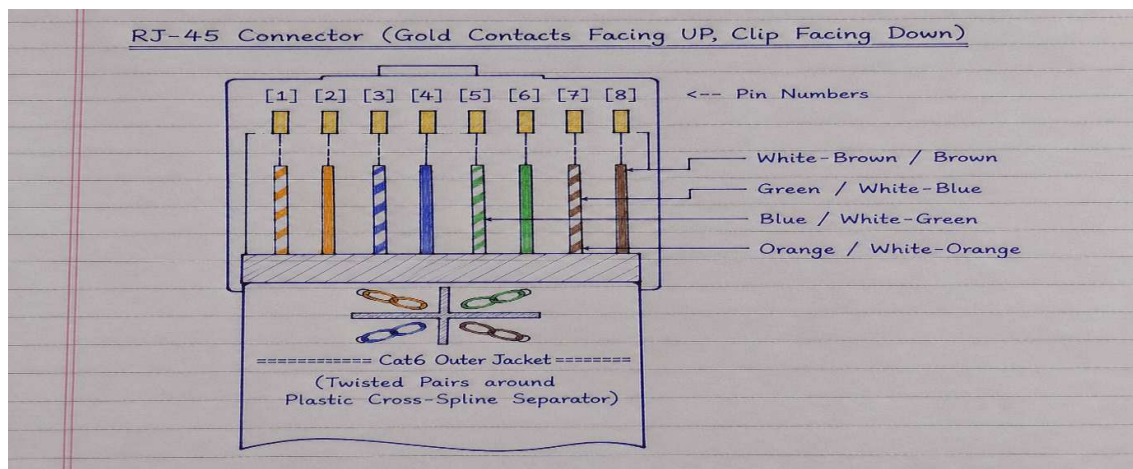
Objective: Represent the RJ-45 connector pinout diagram and provide the exact color coding sequence for both T568A and T568B Ethernet wiring standards.

1. T568A vs T568B Color Code Comparison Table:

Pin Number	T568A Color Code	T568B Color Code	Pin Signal Function
Pin 1	White-Green	White-Orange	Transmit Data (+)
Pin 2	Green	Orange	Transmit Data (-)
Pin 3	White-Orange	White-Green	Receive Data (+)
Pin 4	Blue	Blue	Unused / Gigabit Data

Pin Number	T568A Color Code	T568B Color Code	Pin Signal Function
Pin 5	White-Blue	White-Blue	Unused / Gigabit Data
Pin 6	Orange	Green	Receive Data (-)
Pin 7	White-Brown	White-Brown	Unused / Gigabit Data
Pin 8	Brown	Brown	Unused / Gigabit Data

2. Text-Based RJ-45 Pin Layout Diagram:



Task 2:

Comparison of Key Ethernet Standards Specifications

Specification Feature	10BASE-T	100BASE-TX	1000BASE-T
Common Name	Standard Ethernet	Fast Ethernet	Gigabit Ethernet
Data Transfer Speed	10 Mbps	100 Mbps	1 Gbps (1000 Mbps)
Supported Cable Type	Cat3 / Cat5 UTP	Cat5 / Cat5e UTP	Cat5e / Cat6 UTP
Maximum Cable Length	100 meters (328 ft)	100 meters (328 ft)	100 meters (328 ft)
Connector Type	RJ-45	RJ-45	RJ-45
Wire Pairs Used	2 Pairs (4 Wires)	2 Pairs (4 Wires)	4 Pairs (8 Wires)

Task 3:

Laptop-to-Laptop Direct Ethernet Connection Scenario

Objective: Select the appropriate Ethernet cable type for connecting two host laptops directly without an intermediary network switch and explain the underlying signaling logic.

1. Cable Recommendation:

- **Traditional/Standard Answer: Ethernet Crossover Cable**
 - **Modern Hardware Answer: Straight-Through Cable (utilizing Auto-MDI/MDIX feature)**
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2. Technical Explanation & Wiring Reasoning:

1. Transmit and Receive Pin Mapping (The Wiring Logic):

- Network Interface Cards (NICs) on identical devices (like two laptops) transmit data on **Pins 1 & 2 (Tx)** and receive data on **Pins 3 & 6 (Rx)**.
- If a **Straight-Through cable** is used without Auto-MDIX, Laptop 1's Transmit pins (1 & 2) connect directly to Laptop 2's Transmit pins (1 & 2), resulting in a signal collision and zero communication.
- A **Crossover Cable** swaps the wire pairs at one end (one end **T568A**, the other end **T568B**). This maps **Laptop 1 Tx (Pins 1 & 2) → Laptop 2 Rx (Pins 3 & 6)** and vice versa, enabling seamless bidirectional data transfer.

2. Impact of Modern Auto-MDI/MDIX Technology:

- Almost all modern laptops and Gigabit Ethernet adapters (1000BASE-T) feature **Auto-MDI/MDIX** (Automatic Medium Dependent Interface

Crossover). This technology automatically detects the connected cable type and electronically flips the Tx/Rx pin pairs internally, allowing a standard **Straight-Through cable** to work for direct laptop-to-laptop connection.

Task 4:

Real-World Ethernet Devices & Supported Standards

Objective: Identify three common networking devices used in a home or college environment and state their corresponding supported Ethernet standards.

1. Home Wi-Fi Router / Fiber ONU Broadband Modem

- **Device Type:** Primary Home/Office Gateway Router
- **Likely Ethernet Standard: 1000BASE-T (Gigabit Ethernet)**
- **Speed Supported:** Up to 1 Gbps (1000 Mbps)
- **Reasoning:** Modern fiber broadband routers require Gigabit WAN/LAN ports to deliver high-speed ISP internet connections (300 Mbps to 1 Gbps) across the home network.

2. College Computer Lab Desktop Workstation (PC)

- **Device Type:** Enterprise/Academic Client PC

- **Likely Ethernet Standard: 1000BASE-T (Gigabit Ethernet)**
- **Speed Supported:** Up to 1 Gbps (1000 Mbps)
- **Reasoning:** Modern motherboard Network Interface Cards (NICs) come built-in with Gigabit Ethernet ports to support fast local server file transfers and software installations in college labs.

3. Network Office Printer / Smart TV

- **Device Type:** Shared Peripheral Device
- **Likely Ethernet Standard: 100BASE-TX (Fast Ethernet) or 1000BASE-T**
- **Speed Supported:** Up to 100 Mbps (Fast Ethernet)
- **Reasoning:** Printers and smart TVs do not require multi-gigabit throughput for daily tasks like receiving print queues or streaming 4K video, making 100BASE-TX fully sufficient for their operational needs.

